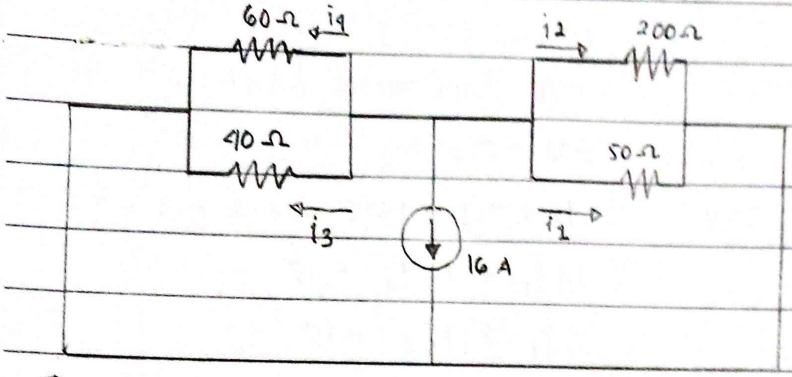
Tsaniya Faridah Maulida
NIM: 211331031
kelas: 1A - TTE

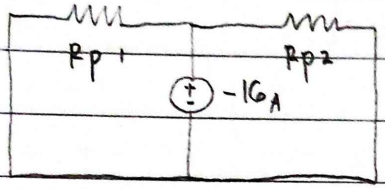
UTS
Rangkaian listrik

Rabu, 10 November 2021
No. :

1. Tentukan besar I_1, I_2, I_3 dan I_4!



Jawab:



$$R_{p1} = \frac{60 \times 40}{60 + 40} = \frac{2400}{100} = 24\ \Omega$$

$$R_{p2} = \frac{200 \times 50}{200 + 50} = \frac{10000}{250} = 40\ \Omega$$

$i_S = -16\ A$

$$i_{Rp1} = \frac{R_{p2}}{R_{p1} + R_{p2}} \times i_S = \frac{40\ \Omega}{24\ \Omega + 40\ \Omega} \times -16A = \frac{40\ \Omega}{64\ \Omega} \times -16A = -10A$$

$$i_{Rp2} = \frac{R_{p1}}{R_{p1} + R_{p2}} \times i_S = \frac{24\ \Omega}{24\ \Omega + 40\ \Omega} \times -16\ A = \frac{24\ \Omega}{64\ \Omega} \times -16A = -6A$$

$$i_4 = \frac{R_3}{R_3 + R_4} \times i_{Rp1} = \frac{40\ \Omega}{40\ \Omega + 60\ \Omega} \times -10A = \frac{40\ \Omega}{100\ \Omega} \times -10A = -4A$$

$$i_1 = \frac{R_2}{R_1 + R_2} \times i_{Rp1} = \frac{200\ \Omega}{50\ \Omega + 200\ \Omega} \times -6A = \frac{200\ \Omega}{250\ \Omega} \times -6A = -4,8\ A$$

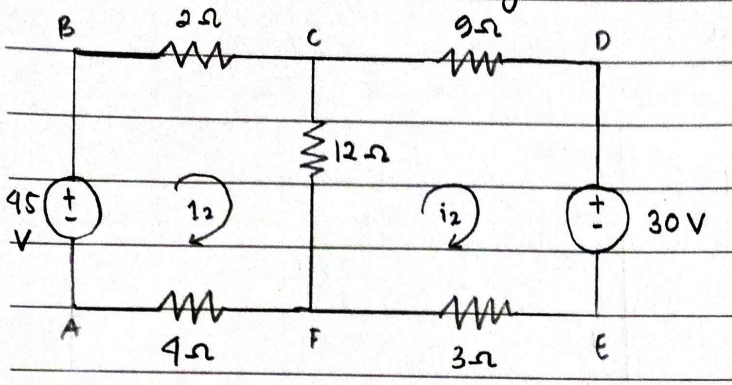
$$i_3 = \frac{R_4}{R_3 + R_4} \times i_{Rp1} = \frac{60\ \Omega}{40\ \Omega + 60\ \Omega} \times -10A = -6\ A$$

$$i_2 = \frac{R_1}{R_2 + R_1} \times i_{Rp1} = \frac{50\ \Omega}{200\ \Omega + 50\ \Omega} \times -6A = -1,2\ A$$

180 mm x 257 mm
KIKY

No. :

2. Tentukan i_1 dan i_2 dengan analisa Mesh !



KVL pada mesh ABCFA

$\Sigma V = 0$

$-95 + 2i1 + 12i1 - 12i2 + 9i2 = 0$

$18I_1 - 12I_2 = 45$

$6I_1 - 9I_2 = 15$

KVL Pada Mesh CDEF

$\Sigma V : 0$

$-30 + 9I_2 + 12I_2 - 12I_1 + 3I_2 = 0$

$-12I_1 + 24I_2 = 30$

$-2I_1 + 9I_2 = 5$

Penyelesaian 1 dan 2

$6I_1 - 9I_2 = 15$

$-2I_1 + 9I_2 = 15$ +

$9I_1 = 20$

$I_1 = \frac{20}{9} = 5$ //

$6(5) - 9I_2 = 15$

$30 - 9I_2 = 15$

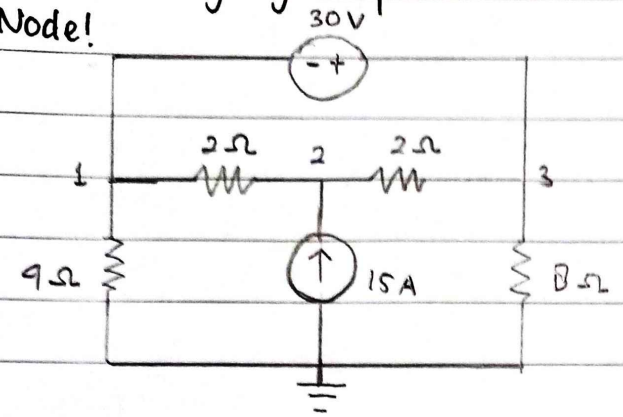
$-9I_2 = 15 - 30$

$I_2 = \frac{-15}{-9} = 3{,}75$ //

KIKY

No. :

3. Tentukan tegangan pada Node 1, 2, dan 3 dengan menggunakan analisa Node!



$I_1 + I_2 = 15$

$\frac{V_2 - V_1}{2} + \frac{V_2 - V_3}{2} = 15$

$V_2 - V_1 + V_2 - V_3 = 30$

$-V_1 + 2V_2 - V_3 = 30 \quad \ldots (1)$

$I_1 + I_2 = \frac{V_1 - 0}{9} + \frac{V_3 - 0}{8}$

$\frac{V_2 - V_1}{2} + \frac{V_2 - V_3}{2} = \frac{V_1 - 0}{9} + \frac{V_3 - 0}{8}$

$9V_2 - 9V_1 + 9V_2 - 9V_3 = 2V_1 + V_3$

$-6V_1 + 8V_2 - 5V_3 = 0 \quad \ldots (2)$

$-9V_1 + 8V_2 - 9V_3 = 120$

$-6V_1 + 8V_2 - 5V_3 = 0 \quad -$

$2V_1 + V_3 = 120 \quad \ldots (3)$

$-V_1 - 30 + V_3 = 0$

$-V_1 + V_3 = 30 \quad \ldots (4)$

- Eliminasi (4) dan (3)

$2V_1 + V_3 = 120$

$-V_1 + V_3 = 30 \quad -$

$3V_1 = 90$

$V_1 = 30\ V$

$-V_1 + V_3 = 30$

$-30 + V_3 = 30$

$V_3 = 60\ V$

$-V_1 + 2V_2 - V_3 = 30$

$-30 + 2V_2 - 60 = 30$

$2V_2 = 120$

$V_2 = 60$

Node 1

$5 = 3 + \frac{V_1 - V_2}{2} + \frac{V_1 - V_2}{8} + \frac{V_1}{9}$

$V_1 - V_3 + \frac{V_1 - V_2}{9} + \frac{V_1}{2} = 9$

$7V_1 - V_2 - 9V_3 = 16 \quad \ldots (1)$

Node 2

$\frac{V_1 - V_2}{8} = \frac{V_2}{2} + \frac{V_2 - V_3}{9}$

$V_1 - V_2 = 9V_2 + 2V_2 - 2V_3$

$V_1 - 7V_2 + 2V_3 = 0 \quad \ldots (2)$

180 mm x 257 mm
KIKY

No. :

Node 3

$$3 + \frac{12 - V_3}{8} + \frac{V_1 - V_3}{2} + \frac{V_2 - V_3}{4} = 0$$

$$24 + 12 - V_3 + 4V_1 - 4V_3 + 2V_2 - 2V_3 = 0$$

$$4V_1 + 2V_2 - 7V_3 = -36 \quad \ldots (3)$$

$$\begin{bmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 16 \\ 0 \\ -36 \end{bmatrix}$$

$\Rightarrow AV = B$

$V = A^{-1} B$

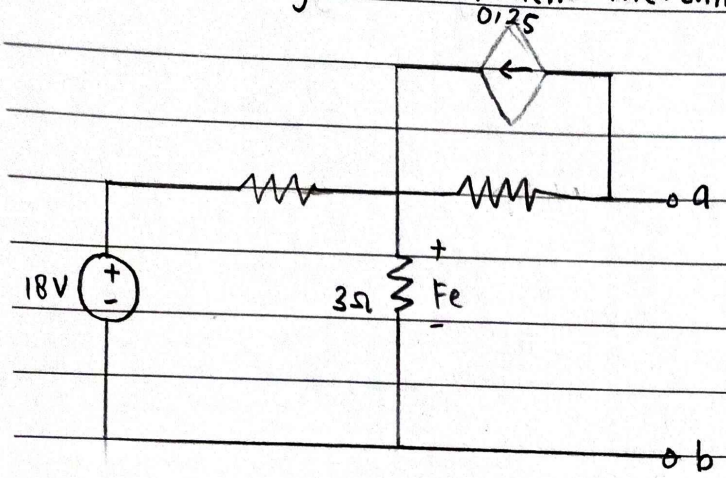
$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 7 & -1 & -4 \\ 1 & -7 & 2 \\ 4 & 2 & -7 \end{bmatrix}^{-1} \begin{bmatrix} 16 \\ 0 \\ -36 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 4.933 \\ 12.267 \end{bmatrix}$$

$\therefore V_1 = 10V, \quad V_2 = 4.933, \quad V_3 = 12.267\ V$

No. :

4. Tentukan Rangkaian Ekivalen Thevenin dan Norton pada terminal a-b !



$(1 + (-i_2) + i_3 + 0{,}25\ V_0 = 0$

$\frac{18 - V_0}{6} - \frac{V_0}{3} + \frac{V_a - V_0}{2} + 0{,}25\ V_0 = 0$

$18 - V_0 - 2V_0 + 3V_a - 3V_0 + 1{,}5\ V_0 = 0$

$-9{,}5\ V_0 + 3V_a = -18 \quad \ldots (1)$

$-i_3 - 0{,}25\ V_0 = 0$

$\frac{V_0 - V_a}{2} - 0{,}25\ V_0 = 0$

$V_0 - V_a - 0{,}5\ V_0 = 0$

$0{,}5\ V_0 - V_a = 0 \quad \ldots (2)$

Eliminasi 1 & 2

$-9{,}5\ V_0 + 3V_a = -18 \mid \times 1$

$0{,}5\ V_0 - V_a = 0 \mid \times 9$

$-9{,}5\ V_0 + 3V_a = -18$

$9{,}5\ V_0 - 9V_a = 0$ (+)

$-6V_a = -18$

$V_a = 3V$

Subtitusi ke (2)

$0{,}5\ V_0 - V_a = 0$

$0{,}5\ V_0 - 3 = 0$

$0{,}5\ V_0 = 3$

$V_0 = 6\ V$

KIKY

No. :

$V_{TH} = V_a = 3V$

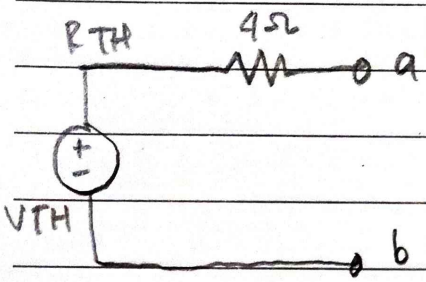
$R_{TH} = (6\Omega \,//\, 3\Omega) + 2\Omega$

$= 2\Omega + 2\Omega$

$= 4\Omega$

$I_N = \frac{V_{TH}}{R_{TH}} = \frac{3}{4} = 0{,}75A$

Thevenin



Norton

